

PROCESS ANALYSIS REPORT

# Sample Output Report

Generated:      Format: Wide Format      Transactions: 425

## 1. DATA QUALITY & COVERAGE

No major data quality issues detected.

| METRIC                                | VALUE      |
|---------------------------------------|------------|
| Rows in file                          | 425        |
| Transactions parsed (100% parse rate) | 425        |
| Complete transactions                 | 425 (100%) |
| Partial transactions                  | 0          |
| Excluded rows                         | 0          |
| Out-of-order anomalies                | 0          |
| Rework transactions                   | 0          |
| Outliers (>3σ)                        | 6          |

## 2. CORE CYCLE TIME METRICS

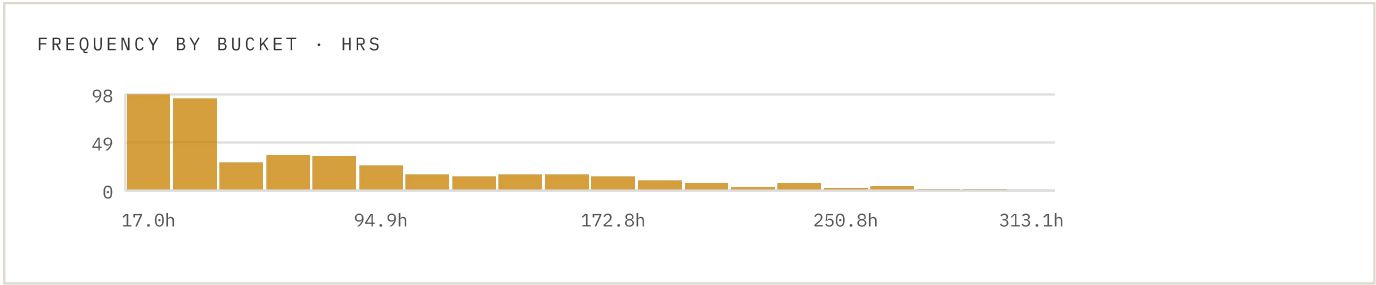
|                                                                     |                                                                 |                                                                              |                                                                                             |
|---------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <div>MEAN CYCLE TIME</div> <div>83.5h</div> <div>median 59.8h</div> | <div>STD DEVIATION</div> <div>64.2h</div> <div>P90 179.9h</div> | <div>VARIABILITY (CV)</div> <div>77%</div> <div>min 17.0h · max 328.7h</div> | <div>OBSERVED VA RATIO</div> <div>100.0%</div> <div>avg VA% per txn <a href="#">Δ</a></div> |
|---------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|

**Observed VA Ratio note:** Ratio is 100.0% because all mapped steps are typed as VA. This reflects your step type configuration in Step 1, not a measurement of actual value-add time. Reclassify any waiting or non-value-add steps to produce a meaningful ratio.

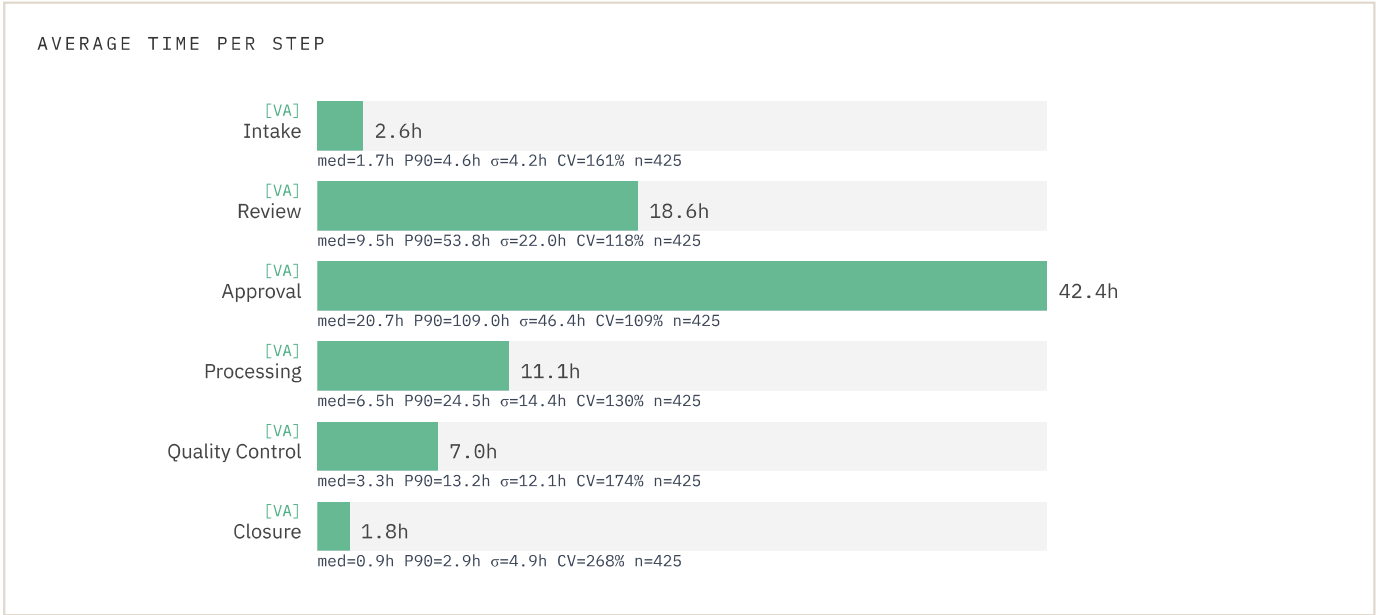
## 3. ANALYSIS WINDOWS

| SETTING                       | VALUE                                                |
|-------------------------------|------------------------------------------------------|
| Date Basis                    | Process Start Date                                   |
| Current-State Window (Cp/Cpk) | 2024-10-15 → 2025-03-15 · <b>152 transactions</b>    |
| Historical Window (Pp/Ppk)    | 2024-01-02 → 2026-05-12 · <b>425 transactions</b>    |
| Sigma tied to                 | Current-State Window (threshold-based defect metric) |

4. CYCLE TIME DISTRIBUTION



5. STEP-LEVEL ANALYSIS



| STEP            | TYPE | AVG   | MEDIAN | P90    | STD DEV | CV   | COVERAGE (N/TOTAL) |
|-----------------|------|-------|--------|--------|---------|------|--------------------|
| Intake          | VA   | 2.6h  | 1.7h   | 4.6h   | 4.2h    | 161% | 425 / 425 (100%)   |
| Review          | VA   | 18.6h | 9.5h   | 53.8h  | 22.0h   | 118% | 425 / 425 (100%)   |
| Approval        | VA   | 42.4h | 20.7h  | 109.0h | 46.4h   | 109% | 425 / 425 (100%)   |
| Processing      | VA   | 11.1h | 6.5h   | 24.5h  | 14.4h   | 130% | 425 / 425 (100%)   |
| Quality Control | VA   | 7.0h  | 3.3h   | 13.2h  | 12.1h   | 174% | 425 / 425 (100%)   |
| Closure         | VA   | 1.8h  | 0.9h   | 2.9h   | 4.9h    | 268% | 425 / 425 (100%)   |

6. SPECIFICATION ASSUMPTIONS & SIX SIGMA

SPECIFICATION ASSUMPTIONS

USL:       **88 hrs**

LSL:       **20 hrs**

*All sigma and capability values are conditional on these thresholds.*

Threshold-based defect metric — sigma reflects performance relative to the USL above, calculated from the Current-State Window only.

SIGMA LEVEL

2.72σ

Below average

DPMO

111,842

DEFECTS / TOTAL

17 / 152

11.2%

7. PROCESS CAPABILITY

CURRENT-STATE CAPABILITY · N=152

CP

0.436

Potential

CPK

0.351

Centered

HISTORICAL PERFORMANCE · N=425

PP

0.176

Potential

PPK

0.023

Centered

Critically incapable — current-state requires immediate action

Poor historical performance — process has not been capable over this period

Centering (K)

20%

8. DISTRIBUTION DIAGNOSTICS & NON-NORMAL CAPABILITY

Moderate non-normality — JB=125.0, p=0.000 (normality rejected)

Best-fit Lognormal and rejected normality (p=0.000) suggest classical Cpk may mischaracterize tail behavior — the fitted CDF provides a more accurate estimate of defect probability.

| DISTRIBUTION | LOG-LIK | AIC    | ΔAIC  | BIC    |
|--------------|---------|--------|-------|--------|
| Lognormal ★  | -3977.4 | 7958.7 | 0.0   | 7966.8 |
| Gamma        | -3998.0 | 8000.0 | 41.2  | 8008.1 |
| Weibull      | -4009.1 | 8022.3 | 63.5  | 8030.4 |
| Normal       | -4112.3 | 8228.5 | 269.8 | 8236.6 |

CAPABILITY COMPARISON

|                                                        |                              |
|--------------------------------------------------------|------------------------------|
| Classical Cpk — full fitted sample, n=425 (Estimated)  | 0.023 [-0.008, 0.055] 95% CI |
| Generalized Cpk — Lognormal tail-probability (Modeled) | 0.095                        |
| Fitted defect probability                              | 38.77% [34.3–43.5%]          |
| P(Classical Cpk ≥ 1.33)                                | 0.0%                         |

Note: Classical Cpk here uses the full filtered sample (n=425) required for distribution fitting. The windowed Cpk in §7 uses the Current-State Window only — these will differ when date windows are configured.

⚠ Mixture behavior suspected (BC=0.655). Two-component fit: 41% group (mean 32.7h) and 59% group (mean 119.5h). Stratify on mapped attributes before treating as confirmed.

Fitted parameters and generalized Cpk are Modeled estimates (distributional assumptions apply). Classical Cpk CI uses Kushler-Hurley approximation. Defect probability CI uses exact Beta posterior.

9. DRIVER ANALYSIS

Baseline mean: 83.5h · Overall defect rate: 35.5%

Product\_Type "Enterprise" mean is +14.3% above the overall average

Product\_Type

↳ Product\_Type "Enterprise" mean is +14.3% above the overall average

| GROUP      | N   | MEAN  | VS BASELINE | DEFECT % | CPK   | RELIABILITY |
|------------|-----|-------|-------------|----------|-------|-------------|
| Enterprise | 148 | 95.5h | +14.3%      | 41.9%    | -0.03 | RELIABLE    |
| Premium    | 139 | 75.6h | -9.5%       | 31.7%    | 0.07  | RELIABLE    |
| Standard   | 138 | 78.7h | -5.8%       | 32.6%    | 0.05  | RELIABLE    |

Region

| GROUP     | N   | MEAN  | VS BASELINE | DEFECT % | CPK  | RELIABILITY |
|-----------|-----|-------|-------------|----------|------|-------------|
| West      | 105 | 79.5h | -4.8%       | 36.2%    | 0.05 | RELIABLE    |
| Southeast | 100 | 86.5h | +3.6%       | 36.0%    | 0.01 | RELIABLE    |
| Northeast | 95  | 85.0h | +1.8%       | 37.9%    | 0.02 | RELIABLE    |
| Midwest   | 125 | 83.3h | -0.2%       | 32.8%    | 0.02 | RELIABLE    |

Mean, median, P90, and defect rate are Observed (computed directly from transactions). Cpk is Estimated (classical short-term capability, n ≥ 15 only). vs Baseline = % delta from overall mean; positive = slower.

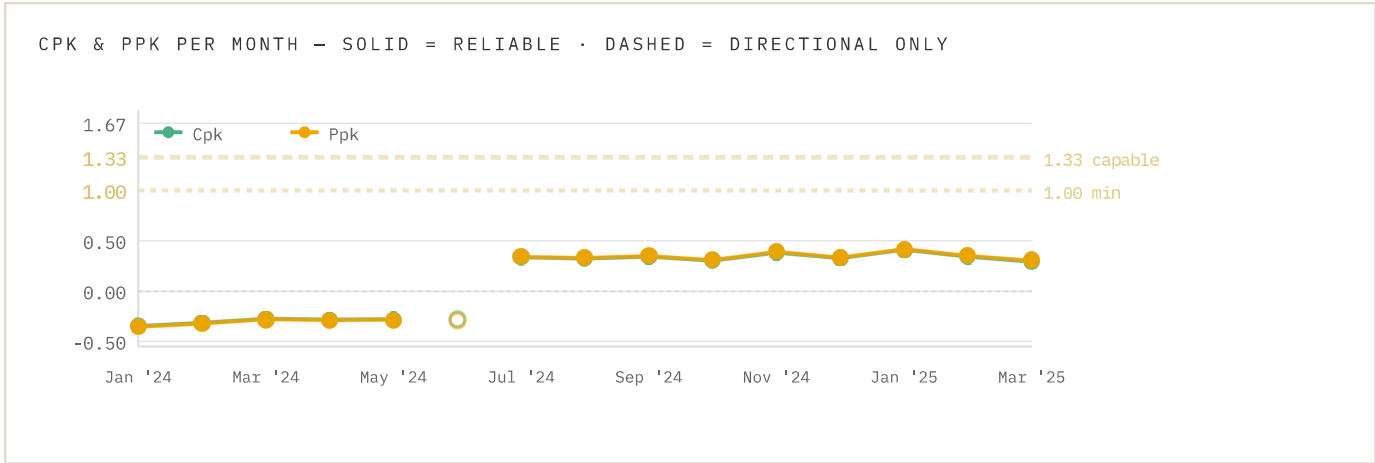
10. CAPABILITY TREND OVER TIME

IMPROVING

Cpk improving — from -0.35 to 0.30 across 14 reliable buckets.

1 directional only · 14 of 15 reliable · Trend driven by Cpk · Monthly bucketing · Min 15 per bucket

Based on the currently filtered transaction set.



KALMAN DRIFT ESTIMATE (Modeled)

|                       |                |
|-----------------------|----------------|
| Drift status          | Stable         |
| Smoothed Cpk          | 0.388          |
| Velocity (per bucket) | 0.0224 ±0.0536 |

Cpk is stable at approximately 0.39 — no significant drift detected.

| FORECAST   | CAPABILITY | 95% BAND |
|------------|------------|----------|
| +1 bucket  | 0.410      | ±0.336   |
| +2 buckets | 0.432      | ±0.439   |
| +3 buckets | 0.455      | ±0.550   |

Kalman smoothing uses a constant-velocity state-space model. Forecast uncertainty widens with each step. Use for trend direction only — not as a point prediction.

11. OUTLIER TRANSACTIONS

Transactions exceeding 3σ above the mean. Review before drawing conclusions.

| TRANSACTION ID | CYCLE TIME |
|----------------|------------|
| TXN-0073       | 284.7h     |
| TXN-0087       | 280.5h     |
| TXN-0118       | 328.7h     |
| TXN-0138       | 279.4h     |
| TXN-0144       | 278.9h     |
| TXN-0148       | 305.0h     |

12. ANALYSIS & RECOMMENDATIONS

1. EXECUTIVE SUMMARY

The process is critically incapable, with a current-state Cpk of 0.351, indicating a need for immediate action. The trend direction is stable, with no significant drift detected. The strongest driver signal is the "Enterprise" product type, which has a mean cycle time 14.3% above the overall average.

2. CAPABILITY CONTEXT & CHANGE OVER TIME

The process is mainly limited by centering, as Cp (0.436) materially exceeds Cpk (0.351), indicating that reducing mean cycle time will yield faster gains than reducing variance. The current-state and historical windows describe materially different processes, with Cpk (0.351) and Ppk (0.023) diverging by more than 0.1, implying significant process change over the measurement period. The trend direction implies that the process is not heading toward capability, with a stable Cpk of approximately 0.39, below the 1.00 minimum threshold.

3. OBSERVED FINDINGS

The data shows a mean cycle time of 83.5 hours, with a median of 59.8 hours, P90 of 179.9 hours, and a coefficient of variation (CV) of 77%. The current-state Cpk is 0.351, indicating critical incapability. Normality is rejected, with a best-fit lognormal distribution, and the generalized Cpk (0.095) is lower than the classical Cpk. The primary bottleneck step is "Approval", with an average of 42.4 hours, compared to the next-longest step "Review" (18.6 hours). The highest-variability step is "Closure", with a CV of 268%. The strongest driver segment is "Enterprise", with a mean cycle time 14.3% above the overall average. The trend is stable, with no significant drift detected. The difference between Cp/Cpk and Pp/Ppk reveals significant process change over time.

4. PROCESS DRIVERS & BOTTLENECKS

The "Approval" step should be targeted first to reduce mean cycle time, as it is the primary constraint. The "Closure" step should be targeted to reduce variability, as it has the highest CV. The strongest driver segment signal implies concentrated performance differences in the "Enterprise" product type. Outlier transactions are not concentrated in a specific segment, suggesting a systemic issue.

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## 5. DISTRIBUTION & CAPABILITY VALIDITY

The normality assumption is not defensible, with a JB result of 125.0 ( $p=0.000$ ), indicating moderate non-normality. The best-fit lognormal distribution changes the capability conclusion materially, with a generalized Cpk (0.095) lower than the classical Cpk. The mixture signal may correspond to an already-identified driver, but requires validation. The reader should use the generalized Cpk, as it is based on the best-fit distribution and provides a more accurate representation of process capability.

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## 6. WORKING HYPOTHESES

- One possible explanation is that the high average cycle time of the "Approval" step (42.4 hours) is due to inefficiencies in the approval process, which could be addressed by streamlining the process or increasing resources.
  - This may indicate that the "Enterprise" product type has unique requirements or complexities that contribute to its higher cycle time, which could be addressed by tailored process improvements.
  - This pattern is consistent with the Cpk/Ppk divergence, suggesting that process changes over time have not been effective in improving capability, and requires validation of the underlying causes.
  - The mixture behavior suspected in the data may correspond to differences in process performance between product types, which could be investigated by stratifying on mapped attributes.
  - The stable trend direction may indicate that improvements are not being sustained over time, which could be due to lack of monitoring or inadequate process controls.
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## 7. RECOMMENDED INVESTIGATIONS

- For hypothesis 1, examine the approval process to identify inefficiencies and opportunities for streamlining, by comparing cycle times for different approval types and analyzing the distribution of approval times.
  - For hypothesis 2, compare the cycle times and defect rates for the "Enterprise" product type to other product types, to determine if the differences are due to unique requirements or complexities.
  - For hypothesis 3, pull data on process changes over time and compare the resulting Cpk values to determine if the changes have been effective in improving capability.
  - For hypothesis 4, examine the distribution of cycle times for each product type to determine if the mixture behavior is due to differences in process performance between product types.
  - For hypothesis 5, compare the trend direction for different product types to determine if the stable trend is due to lack of monitoring or inadequate process controls.
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## 8. SUGGESTED NEXT STEPS

- Implement a streamlined approval process for the "Approval" step, with a goal of reducing average cycle time by 20%.
- Develop tailored process improvements for the "Enterprise" product type, including additional training or resources to address unique requirements or complexities.
- Establish a monitoring system to track the Cpk value weekly, with a reversal threshold of 0.4, to ensure that improvements are sustained over time.
- Conduct regular reviews of process changes to ensure that they are effective in improving capability and addressing the root causes of process variability.
- Develop a plan to address the highest-variability step, "Closure", by implementing process improvements or additional training to reduce the CV.

APPENDIX: METHODS & DEFINITIONS

CYCLE TIME COMPUTATION

**Wide format:** Process start → final step completion. If no start date, Step 1 → final step. Partial rows accepted when final step is valid.

**Event log format:** Events grouped by ID. Latest matching status per step (supports rework). First mapped step → last mapped step.

ANALYSIS WINDOWS

Each transaction is assigned a date based on the chosen Date Basis. The Current-State Window filters transactions for Cp/Cpk; the Historical Window filters for Pp/Ppk. Sigma is also tied to the Current-State Window. When both windows span the same period (or are unconfigured), Cp/Cpk and Pp/Ppk differ only by sample vs population standard deviation.

OBSERVED VA RATIO

Per-transaction: VA intervals ÷ total cycle, capped at 100%. Then averaged. On partial datasets, missing NVA/WAIT intervals inflate this ratio.

SIGMA LEVEL

Defect = cycle time above USL. DPMO = (defects/total) × 1,000,000. Sigma = normsinv(1–DPMO/1M) + 1.5 (long-term with 1.5σ shift). Conditional on USL.

CAPABILITY INDICES

Cp/Cpk: sample stddev (n–1), Current-State Window. Pp/Ppk: population stddev (n), Historical Window. Cp = potential; Cpk = centered capability. Target: Cpk ≥ 1.33.

CAPABILITY TREND

Capability indices are computed per monthly bucket across the filtered transaction set. Trend status (Improving/Stable/Deteriorating) is driven by linear regression slope on Cpk values across reliable buckets. Slope > +0.02 per bucket = Improving; slope < –0.02 = Deteriorating. Minimum 3 reliable buckets (n ≥ 15 each) required for any trend call. Directional buckets (n < 15 but ≥ 8) are shown as indicators only with dashed lines. This trend computation uses the full filtered set with monthly bucketing and is independent of the Analysis Windows configuration.

STEP COVERAGE

Steps marked \* have fewer than 70% of transactions contributing a valid interval. Treat averages as indicative only.